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Small Animal Internal Medicine Oncology

Longitudinal Assessment of QT Apex in Tumor-Bearing Dogs Receiving Doxorubicin Monotherapy

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ABSTRACT

Background: The cardiac morphologic and functional derangements induced by doxorubicin (DOX) are referred to as DOX-induced cardiotoxicity (DOX-IC). Prolongation of the QT apex (QTa) has been identified as a potential marker for the early detection of DOX-IC in humans.

Objectives: Describe changes in QTa that occur in dogs with cancer undergoing DOX monotherapy.

Animals: Forty-five client-owned dogs.

Methods: Descriptive analysis of data routinely recorded as part of case management of dogs with cancer. Dogs included in the study had a confirmed malignant neoplasm, received DOX as a monotherapy, and had at least 4 ECGs > 30 s. All ECGs included in the study were recorded before DOX administration. Five heartbeats with a stable signal and minimal artifact were randomly selected from each dog, and the QTa was blindly evaluated in lead II, lead III, or both. Subsequently, a linear mixed model was used to quantify the effect of a cumulative dose of DOX on the QTa interval, adjusting for the effect of the other clinical variables.

Results: Forty-five dogs met the inclusion criteria. Among them, 39/45 received five DOX treatments and 26/45 received six DOX treatments. For ECG analysis, 234 ECGs were evaluated for changes in the QTa. The average cumulative dose of DOX was 154.1 mg/m^2. There was no significant impact of the DOX treatments on the change in QTa ($p = 0.8$).

Conclusions: Administration of DOX up to a cumulative dose of 154.1 mg/m^2 does not result in QTa prolongation.

1 | Introduction

Doxorubicin (DOX) is a cytotoxic anthracycline drug commonly used in humans and dogs to treat various tumors [1–3]. A variety of mechanisms of action explain its pharmacological effects, including inhibition of topoisomerase IIβ isoforms and free radical generation that can cause myocardial damage, described as DOX-induced cardiotoxicity (DOX-IC) [4–6]. Doxorubicin-induced cardiotoxicity is a substantial dose-limiting factor, categorized into acute

Abbreviations: 2DE, two-dimensional echocardiography; 95%CI, 95% confidence interval; BSA, body surface area; CTRCD, cancer therapeutics-related cardiac dysfunction; DOX, doxorubicin; DOX-IC, doxorubicin-induced cardiotoxicity; ECG, electrocardiography; IM, intact male; IQR, interquartile range; Max, maximum; Min, minimum; NM, neutered male; PD, progressive disease; QTa, QT apex; QTc, corrected QT interval; QTc-B, corrected QT interval by Bazett's formula; QTc-F, corrected QT interval by Fridericia's formula; SD, standard deviation; SF, spayed female.

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and chronic cardiotoxicity. Acute cardiotoxicity is characterized by reversible ECG abnormalities (e.g., repolarization abnormalities), whereas chronic cardiotoxicity is irreversible, dose-dependent, and leads to cardiac dysfunction [7–9]. Doxorubicin-induced cardiotoxicity is well described under the term cancer therapeutics-related cardiac dysfunction (CTRCD) [10]. According to current consensus, CTRCD is defined as a decrease in left ventricular ejection fraction (LVEF) of > 10%, to a value < 53% (normal reference value for two-dimensional echocardiography [2DE], as confirmed by two consecutive echocardiograms 2–3 weeks apart) [10].

In human medicine, endomyocardial biopsy and cardiac imaging are reliable tests to diagnose DOX-IC. Despite its lower sensitivity and specificity compared to endomyocardial biopsy, echocardiography has been favored because of its minimal invasiveness and lower cost-performance ratio [10].

Electrocardiography is considered a practical test for detecting cardiomyopathies. A previous study evaluated the correlation of electrical abnormalities for the early detection of CTRCD [11]. It focused on QT apex (QTa), which was defined as the interval between the start of the Q wave and the peak of the T wave (Figure 1), and concluded that T wave flattening and QTa prolongation may be used as indicators for early detection of DOX-IC [11].

In dogs, many factors (e.g., age, concurrent cardiomyopathies, cumulative dose, and duration of the administration per dose) can affect the onset of DOX-IC [12–14]. The incidence of DOX-IC has been reported to range from 7.4% to 18% as an ECG disturbance, and from 2% to 4% for subsequent congestive heart failure [13–15]. Several ECG abnormalities, including tachycardia, conduction disturbance, and nonspecific ECG changes in R wave, ST segment, or QRS duration, were reported as a result of DOX-IC [14, 16].

The changes in serial ECG analyses during multiple DOX treatments in dogs with cancer have not been described. We hypothesized that QTa changes could be an early indicator of DOX-associated cardiac damage, as in humans. Therefore, our aim was to longitudinally describe QTa changes along with serial treatments of DOX in tumor-bearing dogs.

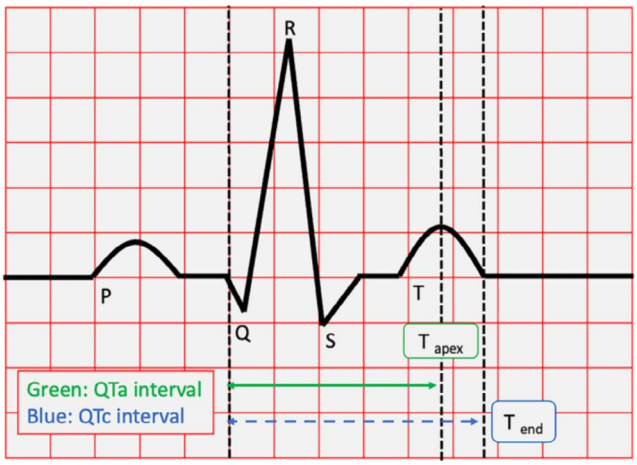

FIGURE 1 | Illustration describing QTa and QTc. The QTa is the interval between the start of the Q wave and the PEAK of the T wave, whereas QTc is the interval between the start of the Q wave and the END of the T wave.

2 | Materials and Methods

2.1 | Study Design and Inclusion Criteria

Ours was a descriptive analysis of data routinely recorded as part of case management of dogs with cancer using the medical record database and ECG library available at the Michigan State University Veterinary Teaching Hospital (MSU-VTH) Oncology Service. The target population was dogs with cancer treated with DOX as monotherapy. The source population consisted of dogs with complete medical records at MSU-VTH. Paired medical records and ECGs from 2011 to 2021 were reviewed, and dogs were included based on the following criteria:

Dogs must have (1) had a malignant neoplasm that is histologically or cytologically confirmed or both, (2) been treated with DOX as a single chemotherapeutic agent, (3) received no previous cytotoxic chemotherapy, (4) had at least 4 ECGs with valid durations > 30 s, and the interval between one ECG recording and another ≤ 4 weeks, (5) had no clinical suspicion for any comorbidities that could affect cardiac function, and (6) had serial medical records that contain signalment, physical examination findings, laboratory data, and treatment information. Dogs that met all inclusion criteria were included in the study population for further analysis.

Signalment (i.e., breed, age, sex), physical examination findings (e.g., presence or absence of cardiac murmur and arrhythmia based on auscultation, and heart rate [HR] and type of neoplasia were documented). Regarding DOX treatment, administration date, dosage (mg/m^2 or mg/kg), total cumulative dose (mg/m^2), and any modification (e.g., dose reduction) from the original DOX protocol also were retrieved.

2.2 | QTa and QTc Measurement

All ECGs were recorded the same day and just before DOX administration. Each ECG was taken with the dog in right lateral recumbency and recorded in the ECG library. All ECGs included in the study were reviewed and verified for QTa measurement by a board-certified cardiologist (R.A.S.) to ensure accuracy and consistency. The QTa was measured as the interval between the start of the Q wave and the peak of the T wave (Figure 1) [11]. Also, the corrected QT interval (QTc) was calculated using Bazett's formula and Fridericia's formula (QTc-B and -F, respectively). Bazett's formula is the QT interval divided by the square root of the RR interval ($QTc\text{-}B = QT/\sqrt{RR}$) [17], whereas Fridericia's formula is the QT interval divided by the cube root of the RR interval ($QTc\text{-}F = QT/\sqrt[3]{RR}$) [18]. The correction was performed in Excel using those specific formulas. The QTa and QTc were measured in lead II, and lead III measurements also were accepted in cases where ECG analysis from lead II was deemed inconclusive. All QTa and QTc (i.e., QTc-B and QTc-F) measurements were performed using five randomly

selected stable complexes with minimal artifact from each recording to ensure accurate QTa measurements, and the average value was used for statistical analysis.

2.3 | DOX Administration

Doxorubicin was administered every 3 weeks as one cycle at a starting dosage of approximately 30 mg/m^2 for dogs weighing ≥15 kg, and 1 mg/kg for dogs weighing <15 kg. The total number of cycles of DOX administered is summarized in Table 1. In the preparation, DOX was diluted with 0.9% sodium chloride. Subsequently, the prepared DOX was administered IV over ≥30 min. The infusion was done by using a long IV catheter placed in a peripheral vein. One dog received DOX at a dosage of 1 mg/kg, although its body weight was >15 kg. The dosage of 1 mg/kg was determined at the discretion of the attending clinician. All dogs received IM administration of diphenhydramine at a dosage of 1–2 mg/kg at least 15 min before DOX administration as a premedication.

2.4 | Comparative Analysis of QTa and QTc Between Two Different DOX Standard Administration Scales

In veterinary medicine, two dosing regimens are conventionally used based on the patient's weight: 30 mg/m^2 for dogs >15 kg and 1 mg/kg for dogs <15 kg [12, 19–21]. To investigate whether the two different dosages of DOX would significantly induce QTa and QTc changes with increasing cumulative dose, we grouped our study population into two groups: large and small breed dogs. Dogs that underwent DOX administration at a 30 mg/m^2 were categorized as large breed, whereas those that received DOX at a 1 mg/kg were categorized as small breed. The QTa and QTc results were compared between the large and small breed dogs at each cycle.

TABLE 1 | Demographic data at baseline.

N	**45**	**Breed**	***n***
Age, median (range), years	9 (1–13)	Mixed breed dog	9
Sex, *n* (%)		Golden Retriever	8
Intact male	2 (4.4%)	Labrador Retriever	6
Neutered male	23 (51%)	American Pit Bull Terrier	1
Spayed female	20 (44%)	Argentine Dogo	1
Body weight, median (range), kg	30.6 (4.9–56.8)	Australian Shepherd	1
Cardiac murmur, *n* (%)		Basset Hound	1
Presence	2 (4.4%)	Bernese Mountain Dog	1
Absence	43 (96%)	Boston Terrier	1
Total number of DOX cycles, *n* (%)		Boxer	1
4	45 (100%)	Cairn Terrier	1
5	39 (87%)	Dachshund	1
6	26 (58%)	Doberman Pinscher	1
DOX dose per cycle, median (range), (mg/m^2)	29.5 (22.0–30.6)	English Springer Spaniel	1
DOX cumulative dose, median (range), (mg/m^2)	154.1 (105.0–178.2)	French Bulldog	1
Tumor type	*n*	German Shepherd Dog	1
Apocrine gland anal sac adenocarcinoma	1	Great Dane	1
Hemangiosarcoma	18	Greyhound	1
Lymphoma	7	Hovawart	1
Mammary gland carcinoma	1	Jack Russell Terrier	1
Nonangiomatous nonlymphomatous sarcoma	5	Labradoodle	1
Osteosarcoma	2	Leonberger	1
Sarcoma	2	Pembroke Welsh Corgi	1
Soft tissue sarcoma	5	Rottweiler	1
Thyroid carcinoma	2	Staffordshire Bull Terrier	1
Transmural mucinous adenocarcinoma	1		
Undifferentiated malignant neoplasm	1		

2.5 | Statistical Analysis

For the demographic data, continuous variables, such as age and body weight, were described using median and range; categorical data (e.g., tumor type and breed) were summarized using percentage. Fisher's exact test was used to compare the demographic characteristics between large and small breed dogs. The Wilcoxon rank sum test was used to compare the differences in each administration scale of DOX (30 mg/m^2 vs. 1 mg/kg).

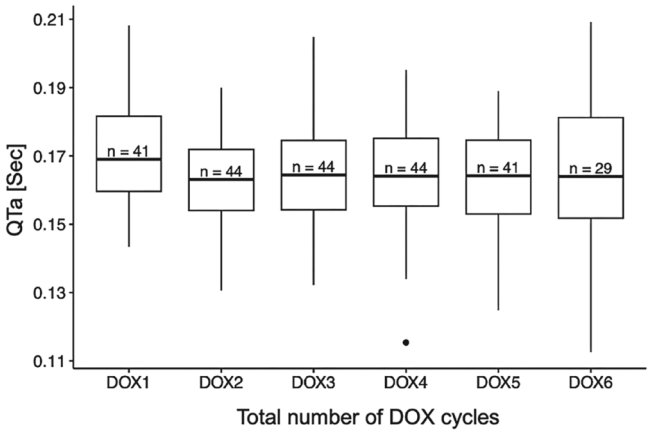

FIGURE 2 | The box and whisker plot showing QTa (s) at each cycle of DOX. Outliers were presented as a black dot.

Multivariable regression analysis was conducted to evaluate the independent effects of clinical variables on QTa and QTc values while accounting for potential confounding factors. A mixed effects linear model was applied, with repeated measures nested within individual dogs to account for the correlation between observations from the same dog. The following factors were included in the model: age (years), sex (intact male, neutered male, spayed female), DOX cycle (1–6 cycles), body size (large vs. small), HR (bpm), presence of a cardiac murmur (present vs. absent), body weight (kg), and body surface area (m^2) and the following formula was used:

$$Y_{ij} = \beta_0 + \beta_1 x_{1ij} + \beta_2 x_{2ij} + \cdots + \beta_m x_{mij} + Z_j + \epsilon_{ij}$$

where Y_{ij} represents the response variable (QTa) for the ith DOX cycle observation of the jth dog. The x_{mij} are the fixed effect predictors (listed above), and $\beta_1, \ldots \beta_m$ are the corresponding fixed-effects coefficients to be estimated. Z_j is the random intercept associated with the jth dog, assumed to follow a normal distribution with mean 0 and variance. The residual error term ϵ_{ij} is assumed to be independent and identically distributed, following a normal distribution with mean 0 and a constant variance. Fixed effect variables were selected based on clinical relevance and prior evidence suggesting their influence on QTa and QTc values. Backward elimination was applied to retain significant predictors. Random effects were included to account for within-dog correlations arising from repeated measures. All analyses

TABLE 2 | The average QTa (s) at each cycle of DOX and its comparison between large and small breed dogs.

			QTa (s)						
DOX cycle	**Group**	***N***	**Median**	**Min**	**Max**	**IQR**	**Mean**	**SD**	***p***
DOX 1	All	41	0.169	0.143	0.208	0.16–0.182	0.17	0.017	0.93
	Large	33	0.169	0.144	0.144	0.16–0.182	0.17	0.014	
	Small	8	0.169	0.143	0.208	0.157–0.186	0.173	0.024	
DOX 2	All	44	0.163	0.131	0.19	0.154–0.172	0.163	0.014	0.44
	Large	36	0.164	0.131	0.19	0.155–0.172	0.163	0.015	
	Small	8	0.158	0.141	0.182	0.151–0.168	0.16	0.014	
DOX 3	All	44	0.164	0.132	0.205	0.154–0.174	0.165	0.017	0.66
	Large	36	0.164	0.132	0.205	0.155–0.175	0.166	0.018	
	Small	8	0.165	0.134	0.187	0.153–0.168	0.161	0.016	
DOX 4	All	44	0.164	0.115	0.195	0.155–0.175	0.163	0.017	0.78
	Large	37	0.165	0.115	0.195	0.156–0.175	0.164	0.017	
	Small	7	0.16	0.142	0.181	0.154–0.174	0.163	0.015	
DOX 5	All	41	0.164	0.125	0.189	0.153–0.175	0.163	0.014	0.99
	Large	33	0.164	0.125	0.189	0.153–0.175	0.162	0.015	
	Small	8	0.164	0.149	0.182	0.157–0.168	0.163	0.01	
DOX 6	All	29	0.164	0.113	0.209	0.152–0.181	0.166	0.021	0.33
	Large	25	0.164	0.113	0.209	0.152–0.176	0.164	0.022	
	Small	4	0.18	0.147	0.188	0.168–0.186	0.174	0.019	

Note: p value was calculated using the Wilcoxon rank sum test.
Abbreviations: IQR, interquartile range; Max, maximum; Min, minimum; SD, standard deviation.

and visualizations were conducted using R (version 4.3.2). A $p < 0.05$ was considered significant.

3 | Results

3.1 | Demographic Data

Overall demographic data at initial presentation is summarized in Table 1. Forty-five dogs met our inclusion criteria. Most were middle-aged, large breed, and spayed or neutered. Twenty-five different breeds were represented. Mixed breed dogs, Golden Retrievers, and Labrador Retrievers were most common, and 22 other breeds were represented by one each. There were 11 different tumor types, with hemangiosarcoma being most common. Doxorubicin was administered at a median dose per cycle of 29.5 mg/m^2 (range: 22.0–30.6 mg/m^2). Dose modification (i.e., dose reduction or dose escalation) was implemented in 17/45 dogs. Of them, one dog needed dose modification three times over the course of DOX treatment, four required it twice, and the remainder (12) needed it once. Two dogs were noted to have cardiac murmurs. The first dog had a grade I/VI bilateral apical systolic heart murmur at initial presentation. The second dog developed a grade II–III/VI left apical systolic murmur after the second cycle of DOX. This murmur persisted throughout subsequent cycles.

TABLE 3 | Regression coefficients and their standard errors from a linear mixed effects model of the effect of DOX cycle, presence or absence of cardiac murmur, size of patient (large, small), sex, age, body weight, body weight surface area, and heart rate on QTa (s).

(a) Full multivariable mixed effects linear model for QTa ($N = 45$)					
Characteristic	**Beta**	**Standard error**	**Statistic**	**95% confidence interval**	***p***
DOX cycle	−0.0002	0.000	−0.543	−0.0010, 0.0005	0.6
Cardiac murmur					
No	—	—	—	—	
Yes	−0.0013	0.004	−0.290	−0.0100, 0.0074	0.8
Size					
Large	—	—	—	—	
Small	0.0039	0.009	0.449	*−0.0137, 0.0216*	0.7
Sex					
Intact male	—	—	—	—	
Neutered male	0.0091	0.007	1.25	−0.0056, 0.0238	0.2
Spayed female	0.0006	0.007	0.075	−0.0143, 0.0154	> 0.9
Age	0.0005	0.001	0.956	−0.0006, 0.0016	0.3
Body weight (kg)	−0.0004	0.002	−0.218	−0.0036, 0.0029	0.8
Body surface area (m^2)	0.0174	0.077	0.225	−0.1383, 0.1730	0.8
Heart rate (bpm)	−0.0004	0.000	−11.5	−0.0005, −0.0003	< 0.001
(b) Selected mixed effects linear model for QTa from backward elimination[b] ($N = 45$)					
Characteristic	**Beta**	**Standard error**	**Statistic**	**95% confidence interval**	***p***
DOX cycle[a]	−0.0001	0.000	−0.296	−0.0009, 0.0006	0.8
Size[a]					
Large	—	—	—	—	
Small	0.0022	0.004	0.599	−0.0053, 0.0097	0.6
Sex					
Intact male	—	—	—	—	
Neutered male	0.0091	0.007	1.25	−0.0056, 0.0238	0.2
Spayed female	0.0006	0.007	0.075	−0.0143, 0.0154	> 0.9
Heart rate (bpm)	−0.0004	0.000	−11.8	−0.0005, −0.0003	< 0.001

[a]DOX treatment and size were forced to be kept in the model.
[b]Variance of the random effect was 7.1311e-05.

3.2 | Longitudinal Evaluation of QTa and QTc

From 45 dogs included in the study, we identified 243 ECG entries for longitudinal QTa/QTc analysis. Eight were removed from our dataset because of inadequate quality or missing data on a given day. Therefore, 235 ECG entries made up the final pool for data analysis.

No statistical difference was found in the QTa value across DOX cycles (Figure 2). The QTa value at each DOX cycle is

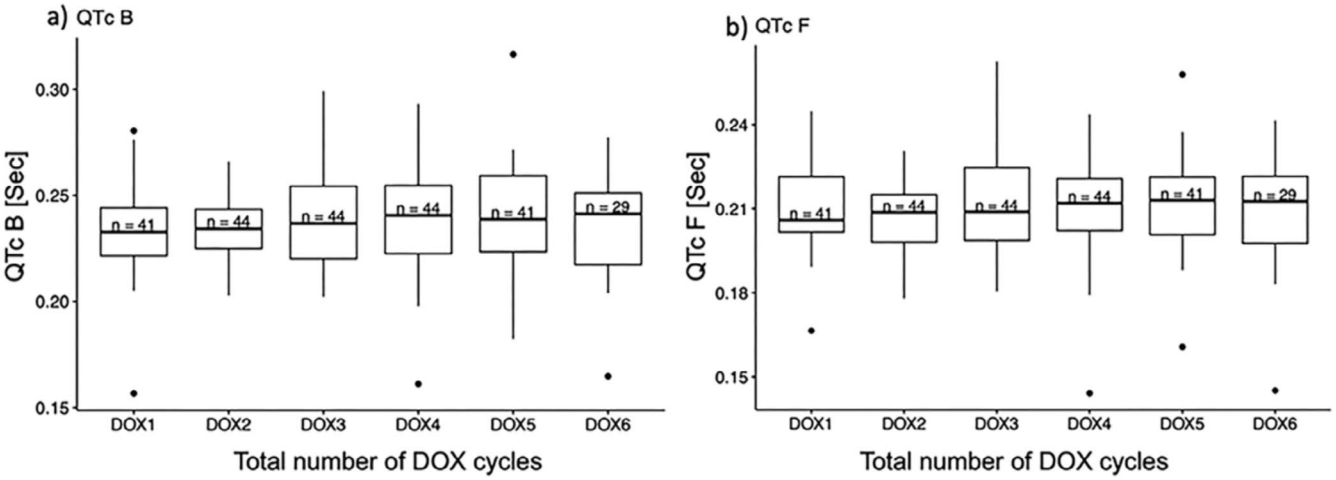

FIGURE 3 | The box plots showing the changes of QTc-B (a) and -F (b) (s) at each cycle of DOX. Outliers were presented as a black dot.

TABLE 4 | Regression coefficients and their standard errors from a linear mixed effects model of the effect of DOX cycle, presence or absence of cardiac murmur, size of patient (large, small), sex, age, bodyweight, body weight surface area, and heart rate on QTc-B and QTc-F (s).

QTc-B (*N* = 45)					
Characteristic	**Beta**	**Standard error**	**Statistic**	**95% confidence interval**	***p***
DOX cycle[a]	0.0005	0.001	0.846	−0.0006, 0.0016	0.4
Size[a]					
Large	—	—	—	—	
Small	0.0051	0.006	0.896	−0.0064, 0.0166	0.4
Sex					
Intact male	—	—	—	—	
Neutered male	0.0017	0.011	0.161	−0.0197, 0.0231	0.9
Spayed female	−0.0111	0.011	−1.04	−0.0327, 0.0105	0.3
Heart rate (bpm)	0.0004	0.000	7.42	0.0003, 0.0005	< 0.001
QTc-F (*N* = 45)					
Characteristic	**Beta**	**Standard error**	**Statistic**	**95% confidence interval**	***p***
DOX cycle[a]	0.0004	0.000	0.789	−0.0005, 0.0013	0.4
Size[a]					
Large	—	—	—	—	
Small	0.0042	0.005	0.866	−0.0056, 0.0139	0.4
Sex					
Intact male	—	—	—	—	
Neutered male	0.0052	0.009	0.581	−0.0129, 0.0234	0.6
Spayed female	−0.0061	0.009	−0.670	−0.0244, 0.0123	0.5

[a]DOX treatment and size were forced to be kept in the model.

summarized in Table 2. In addition, we analyzed longitudinal changes in an individual's QTa (Figure 4).

For the mixed effects linear model, initially, eight factors (age, sex, DOX cycle, body size [large or small], HR, presence or absence of cardiac murmur, body weight, and body surface area) were included in the model. Doxorubicin cycle was not a significant contributor to longitudinal change in QTa ($p = 0.8$; Table 3). However, HR was a significant contributor to QTa, adjusting for other factors ($p < 0.001$; Table 3). Thus, the results indicated that HR significantly contributes to the longitudinal change of QTa, but not DOX cycle. Additionally, in analyzing the QTc-B and -F, no significant difference was found in the QTc value across DOX cycles, regardless of the correction formulas (Figure 3, Table 4).

3.3 | Comparative Analysis of QTa and QTc Between the Two Different Standard Administration Scales

We compared the QTa and QTc values between large and small breed dogs (i.e., those that received DOX at 30 mg/m^2 vs. those that received 1 mg/kg). The demographic data of each cohort are summarized in Table 5. No significant difference in QTa and QTc was found at any DOX cycle (Tables 2 and S1). Additionally, spaghetti plots were developed to describe changes in QTa and QTc over time at the individual dog level (Figure 4, Figure S1).

TABLE 5 | Baseline demographic characteristics of study dogs by large and small breed dogs.

	Large	Small	*p*
N	37	8	—
Age, median (range), years	9 (1–13)	8.5 (4–12)	> 0.9
Sex, *n* (%)			0.6
Intact male	2 (5.4%)	0 (0%)	
Neutered male	20 (54%)	3 (38%)	
Spayed female	15 (41%)	5 (63%)	
Body weight, median (range), kg	33.4 (17.3–56.8)	9.0 (4.9–15.7)	< 0.001
Cardiac murmur, *n* (%)			> 0.9
Presence	2 (5.4%)	0 (0%)	
Absence	35 (95%)	8 (100%)	
Total number of DOX cycles, *n* (%)			
4	37 (100%)	8 (100%)	—
5	32 (87%)	7 (88%)	—
6	22 (60%)	4 (50%)	—

Note: p value was calculated by Fisher's exact test and Wilcoxon rank sum test.

3.4 | Clinical Outcome

The median follow-up time of the studied population from the last cycle of DOX administered and the first hospital visit was 65 days (range: 0–1516 days); 37 of 45 dogs were alive, and eight of 45 dogs died or were euthanized. Of the 37 dogs alive at the last hospital visit, 15 had their tumors under control, nine had progressive disease (PD; suspected based on diagnostic images in six dogs and confirmed by cytological evaluation in three dogs), and five presented to MSU-VTH for other conditions (e.g., development of another type of tumor). The remaining eight were lost to follow-up after completing DOX treatment. No diagnostic imaging or ECG was used during the follow-up period for the sole purpose of evaluating cardiac function. However, clinical signs indicative of DOX-IC were not documented.

4 | Discussion

In dogs with no functional cardiomyopathies, DOX can be safely administered at a cumulative dose of 150–220 mg/m^2 [12, 14, 15]. In humans, the occurrence of DOX-IC was estimated to be 5% at a cumulative dose of 400 mg/m^2, increasing to 16% at a dose of 500 mg/m^2, 26% at a dose of 550 mg/m^2, and 48% at a cumulative dose of 700 mg/m^2 [22]. The median cumulative dose of DOX in our study was 154.1 mg/m^2, which may not have been sufficient to induce preclinical or clinical DOX-IC. However, given that the risk of DOX-IC increases with cumulative dose, it was reasonable to investigate the cumulative effect of DOX on QTa [9, 22].

Longitudinal changes in the individual QTa and QTc of dogs identified no pattern correlating QTa and QTc values with DOX cycles throughout the protocol. A previous study of human patients reported that 4.8% of the study population developed DOX-IC, and QTa prolongation was detected in 45% at the preclinical stage and 60% at the clinical stage, respectively [11]. Also, the median time to detect DOX-IC in that study was 453 days, with 60% of patients requiring over 3 years for diagnosis of DOX-IC [11]. Another study, despite using a higher DOX dose per cycle (1.5 mg/kg), noted that QT prolongation did not constantly occur during ECG observation, but only sporadically [23]. These results suggest that the sensitivity of detecting QT prolongation may be influenced by timing, observation period, or observer error.

Acknowledging the limitation of a small study population and assuming a similar occurrence of DOX-IC and sensitivity to detect a QTa in a preclinical stage of DOX-IC, we decided to track QTa longitudinally in each dog by using spaghetti plots to identify possible cases that would develop a QTa suggestive of a preclinical stage of DOX-IC. We identified a subpopulation exhibiting QTa trends that warranted close monitoring. However, the underlying oncologic condition dictated both the lifespan and follow-up duration (median, 65 days), limiting our ability to determine the clinical relevance of these trends.

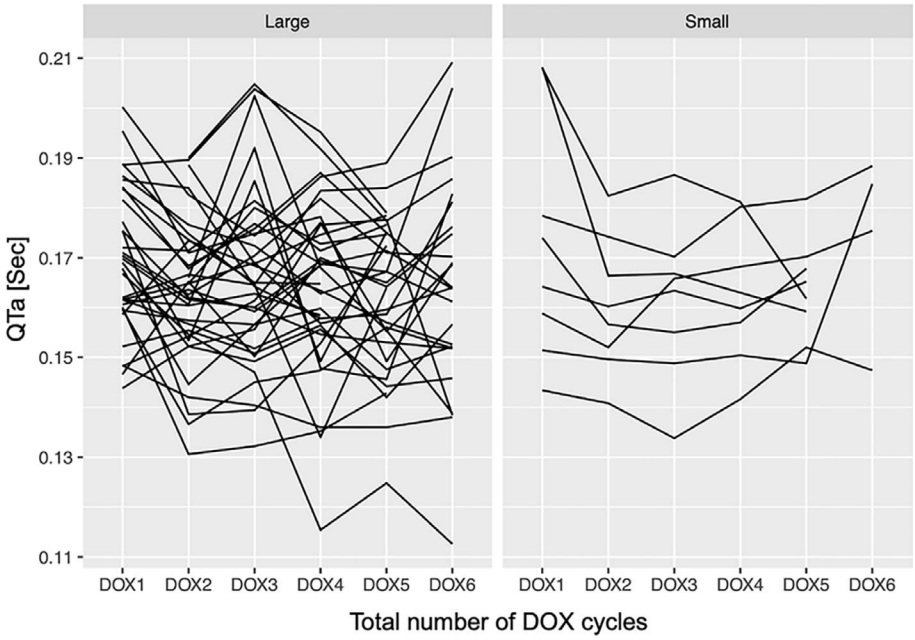

FIGURE 4 | Individual longitudinal changes of QTa, Large and Small breed dogs. This plot shows that individual changes of QTa (s) over time.

When comparing the two currently established administration scales of DOX (large vs. small breed dogs), no significant differences in QTa were observed. A previous study indicated that 20 of 494 (4%) dogs developed clinical DOX-IC, with all cases but one Boston Terrier belonging to the category of large breed dogs [13]. This observation also could be reflected in our study or be a consequence of bias because of the small sample size of a small breed dog population.

Although the specific mechanism of QTa prolongation remains unclear, several factors have been reported. Some studies have shown that QTa prolongation with DOX treatment results from the prolonged action potentials of ventricular cardiomyocytes [24, 25]. Another study indicated that the QTa index in older people is significantly more prolonged than in younger people, suggesting that aging also may impact the QTa interval [26].

In our study, HR was found to be a significant contributor to QTa. Another study documented that increased HR induced by atrial pacing directly affected ventricular repolarization by shortening the QT interval [27]. Because QTa preserves most of the QT interval, an increase in HR seemed to shorten the QTa value, similar to the previous study [27]. However, it remains questionable whether the effect of HR on the QTa is clinically meaningful, because the estimated effect size of HR on QTa is extremely low (−0.0004 s per 1-bpm change) in our model.

The variance of the random effect was 7.1311×10^{-5}, corresponding to a SD of approximately 0.0084 s. Given that the overall median QTa interval in our study ranged from 0.164 to 0.169 s, it indicated relatively low between-dog variability after adjusting for fixed effects. This observation suggests that once clinical variables such as HR and DOX cycle were accounted for, individual baseline differences between dogs contributed minimally to overall QTa variation.

Our study had some limitations. First, the sample size was small. Second, because of its retrospective nature, not all potential factors influencing QTa and QTc were completely assessed. Third, in our study cohort, we could not completely exclude patient comorbidities that could have affected cardiac function (e.g., dilated cardiomyopathy, myxomatous mitral valve degenerative disease, hypothyroidism, electrolyte abnormalities). Although the current standard of care screened out suspicion of these diseases, additional diagnostic tests were not implemented to exclude comorbidities.

A previous study pointed to early oxidative homeostasis and autophagy disruption before noticeable changes in biomarkers (i.e., NT-proBNP) [28]. Studies investigating longitudinal changes of oxidative stress and autophagy markers along with clinical biomarkers, such as NT-proBNP and serum troponin I, echocardiography, and QTa, may provide additional insights into preclinical myocardial damage induced by DOX administration.

In summary, we found that a median cumulative DOX administration of 154.1 mg/m^2 did not affect QTa and QTc.

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Disclosure

Authors declare no off-label use of antimicrobials.

Ethics Statement

This study followed the guidelines approved by the Ethics Committee of Michigan State University Veterinary Teaching Hospital. All dogs in

this study were examined and treated with the owner's written consent. Authors declare human ethics approval was not needed.

Conflicts of Interest

The authors declare no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.